Md.Nahin Alam

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europass

ABOUT ME

AI/ML researcher and graduate student specializing in multimodal AI, computer vision, and natural language processing with applications in medical imaging and under-resourced languages. Currently conducting cutting-edge research at University of Colorado Boulder's Centre for UnlieAI Artificial Intelligence Research Lab, contributing to dataset preprocessing, model training, and publication preparation. Papers under review at premier AI conferences including NeurIPS, AAAI, and ICDE.

Extensive experience in developing state-of-the-art deep learning models using PyTorch, TensorFlow, and Hugging Face Transformers for medical imaging and NLP applications. Proficient in Vision-Language Models including CLIP, ViLT, BLIP, and BioViL-T for radiology benchmark development. Strong background in implementing feature selection pipelines using XGBoost, LightGBM, and ensemble methods for biomarker discovery. Skilled in deploying scalable ML solutions on AWS and Google Cloud Platform with Docker, Kubernetes, and CI/CD pipelines.

Demonstrated track record of publishing research at top-tier conferences and workshops, including NeurIPS and AAAI. Led development of RadMultiBench, a comprehensive multimodal benchmark for evaluating vision-language models in radiology. Contributed to multiple research initiatives in weather forecasting, NLP for Bangla language, and feature selection for medical applications. Experienced in mentoring junior researchers and collaborating with cross-functional teams on machine learning projects.

Committed to advancing AI research and education through teaching and lab instruction roles at North South University. Passionate about developing practical ML applications that address real-world challenges in healthcare, under-resourced language processing, and data-driven decision making. Seeking opportunities to contribute to innovative AI research projects that make meaningful impact.

WORK EXPERIENCE

University of Colorado Boulder

Research Assistant

01/2025 – Current

Conducting cutting-edge research in multimodal AI with applications in medical imaging and natural language processing for under-resourced languages. Collaborating with Centre for UntieAI Artificial Intelligence Research Lab, contributing to dataset preprocessing, model training, and publication preparation. Papers under review at premier AI conferences (NeurIPS, AAAI).

North South University

Lab Instructor

02/2025 – Current

Teaching and conducting hands-on laboratory sessions for AI and Machine Learning courses at undergraduate level. Designing lab exercises, evaluating student performance, and providing mentorship. Developing practical demonstrations using PyTorch, TensorFlow, scikit-learn for computer vision and NLP tasks. Facilitating student understanding of neural networks, deep learning architectures, and model deployment.

Machine Learning Lab, North South University Dhaka

University research assistant

01/2025 – 10/2025

Leading research initiatives in multimodal AI, computer vision, and NLP. Developing state-of-the-art models using PyTorch, TensorFlow, and Hugging

Face Transformers for medical imaging and natural language processing applications. Conducting experiments with Vision-Language Models including CLIP, ViLT, BLIP, and BioViL-T for radiology benchmark development. Publishing research at premier AI conferences (NeurIPS, AAAI, ICDE). Mentoring junior researchers on machine learning projects and deep learning architectures. Implementing feature selection pipelines using XGBoost, LightGBM, and ensemble methods for biomarker discovery.

Cyber Physical Systems Lab, North South University Dhaka

Research Assistant

06/2023 – 06/2024

Conducting research on Internet of Things (IoT), cyber-physical systems, and embedded systems. Developing intelligent sensor networks and real-time monitoring systems using Python, C/C++, and MATLAB. Implementing machine learning algorithms for predictive maintenance and anomaly detection in industrial systems. Designing data acquisition and processing pipelines for sensor data using TensorFlow and scikit-learn. Collaborating on smart city and industrial automation projects. Publishing research findings and presenting at academic conferences.

North South University Dhaka

University teaching assistant

06/2024 – 01/2025

Assisting faculty in delivering undergraduate and graduate level courses in Computer Science and Engineering. Conducting tutorial sessions, grading assignments and examinations, holding office hours for student consultations. Supervising laboratory sessions for programming courses in Python, Java, and C/C++. Providing academic support and mentorship to students on software development projects and machine learning assignments. Facilitating discussion sessions on algorithms, data structures, and artificial intelligence concepts.

The Data Island Dhaka

Technical Specialist

06/2024 – 10/2025

Providing technical expertise in cloud computing, database management, and DevOps practices. Architecting and deploying scalable solutions using AWS, Google Cloud Platform, Docker, and Kubernetes. Implementing CI/CD pipelines with Jenkins and Git for automated deployment. Managing PostgreSQL, MongoDB, and Redis databases for high-performance applications. Developing RESTful APIs and microservices architectures. Optimizing system performance and ensuring reliability of production environments. Collaborating with development teams on technical requirements and best practices.

EDUCATION AND TRAINING

01/01/2020 – 14/04/2024 Dhaka

Bachelor of Science North South University

Specialized in Artificial Intelligence, Machine Learning, Computer Vision, and Deep Learning. Key coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Software Engineering, Neural Networks, Natural Language Processing. Capstone project on multimodal AI for medical imaging. Dean's List recognition for academic excellence.

PUBLICATIONS:

• RadMultiBench: A Multimodal Benchmark Reveals Architectural Failures in Modern Vision-Language Models for Radiology (First Author, submitted to ICDE 2026)
• Validating a Consensus Feature Selection Pipeline for Rheumatoid Arthritis Biomarkers: A Synthetic Data Study (First Author, submitted to ICDE 2026)
• Engineering Spatiotemporal Forecasts at Scale: Dhaka Weather with Machine Learning (Co-author, submitted to ICDE 2026)
• A Comparative Study of Lightweight Hybrid CNN-BiLSTM Architectures for Fine-Grained Bangla Reaction Classification (Co-author, submitted to ICDE 2026)
• Face2Profile: Extracting Semantic Features from Face Images (NeurIPS Workshop)
• Contextual Hate Speech Detection (AAAI Workshop)

KEY PROJECTS:

• RadMultiBench - Multimodal AI benchmark for radiology
• Medical Image Analysis Platform using PyTorch
• NLP Pipeline for under-resourced languages
• Weather Forecasting System with LSTM
• Biomarker Discovery Pipeline using ML

• Real-time Sentiment Analysis System

CERTIFICATIONS:
• FastAI Deep Learning Course
• Stanford Machine Learning Specialization
• Andrew Ng Deep Learning Specialization
• freeCodeCamp Data Science Certification
• Google Cloud Platform Fundamentals

Website https://www.northsouth.edu/ | **Field of study** Computer Science and Engineering | **Final grade** 3.81/4.00 (Summa Cum Laude) | **Level in EQF** EQF level 6